ESSENTIAL SURJECTIVITY OF RESTRICTION TO QUASI-COMPACT OPEN IMMERSIONS ON DERIVED CATEGORIES OF QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The derived pushforward Rj_* along a quasi-compact open immersion preserves quasi-coherence of cohomology sheaves. invoked at proof lines 3273-3275 and 3280-3281. Makes the right-inverse mechanism work within D_QCoh.
- Rj_* is right adjoint to restriction, so restriction composed with Rj_* is the identity on D(O_U). used at proof lines 3273-3274. Provides the lifting of objects from U to X.
- Rj_* preserves bounded-below complexes and, when X is quasi-compact, bounded-above complexes. used at proof lines 3276-3284. Gives the bounded variants.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 3274-3275 and 3280-3281 to keep Rj_* within D_QCoh and within the bounded fragments; the essential-surjectivity conclusions are inside the counted span.

Lemma 0.2. *Let X be a scheme and let $j : U \to X$ be a quasi-compact open immersion. The functors*

$$D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_U) \quad \text{and} \quad D^+_{QCoh}(\mathcal{O}_X) \to D^+_{QCoh}(\mathcal{O}_U)$$

are essentially surjective. If X is quasi-compact, then the functors

$$D^-_{QCoh}(\mathcal{O}_X) \to D^-_{QCoh}(\mathcal{O}_U) \quad \text{and} \quad D^b_{QCoh}(\mathcal{O}_X) \to D^b_{QCoh}(\mathcal{O}_U)$$

are essentially surjective.

Proof. The argument preceding the lemma applies for the first case because Rj_* maps $D_{QCoh}(\mathcal{O}_U)$ into $D_{QCoh}(\mathcal{O}_X)$ by Lemma 08D5. It is clear that Rj_* maps $D^+_{QCoh}(\mathcal{O}_U)$ into $D^+_{QCoh}(\mathcal{O}_X)$ which implies the statement on bounded below complexes. Finally, Lemma 08D5 guarantees that Rj_* maps $D^-_{QCoh}(\mathcal{O}_U)$ into $D^-_{QCoh}(\mathcal{O}_X)$ if X is quasi-compact. Combining these two we obtain the last statement. □

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